

Qatar-1b

R-2152

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_121022 · created 2026-08-12T01:12:53+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456222.74311 +/- 0.00011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.156 +/- 0.011
Transit depth (Rp/Rs)^2	2.42 +/- 0.35 [%]
Orbital Inclination (inc)	84.69 +/- 0.91
Airmass coefficient 1 (a1)	1.218 +/- 0.015
Airmass coefficient 2 (a2)	-0.1107 +/- 0.0061
Scatter in the residuals of the lightcurve fit is	1.23 %
Best Comparison Star	None
Optimal Aperture	2.88
Optimal Annulus	5.96
Transit Duration (day)	0.0723 +/- 0.0043

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.156 ± 0.011 vs 0.1463 (deviation 0.88σ)
Duration fit vs published	0.0723 d vs 0.0692 d (deviation 0.73σ)
Mid-time O–C	0.3 min
Depth SNR	14.1
Residual scatter	1.23 %

Light curve

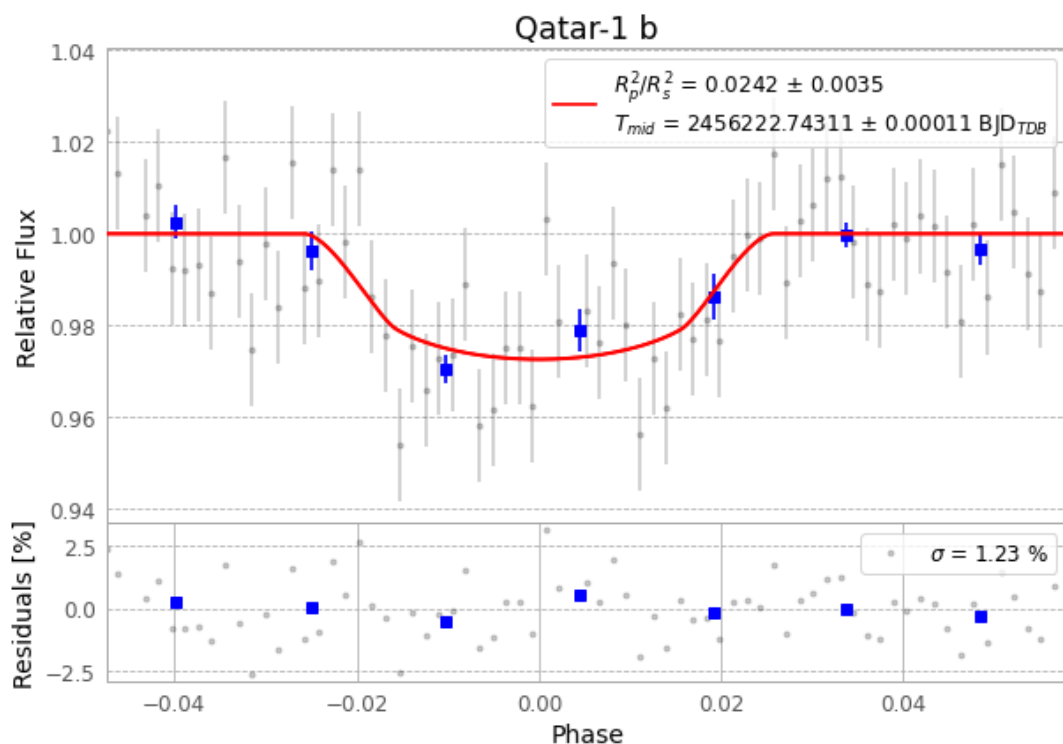


Figure 1 — FinalLightCurve_Qatar-1 b_2012-10-21.png (SHA-256 0f8aefb151740f1c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [350, 253], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a797066b0e650994...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

73 FITS frames (48 MB), Qatar-1121022040943.FITS → Qatar-1121022074512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-121022.log (ab4d79dc8cd1...), FinalLightCurve_Qatar-1 b_2012-10-21.png (0f8aefb15174...), FinalLightCurve_Qatar-1 b_2012-10-21.csv (a1edbdba7c422...)

Compute resources

Wall time 141.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-12 01:12Z.